[Uniqueness
[Thm.
Let ϕ be a real-valued function on $R = [a,b] \times [\alpha, \beta] \subseteq \mathbb{R}^2$.
Consider the initial-value problem

$$\begin{cases} y' = \phi(x,y) \\ y(a) = c \in [\alpha, \beta]. \end{cases}$$

A differentiable function f on $[a,b]$ such that $f(a) = c$ and $\alpha \le f(x) \le \beta$ and

$$f'(x) = \phi(x, f(x)) \quad a \le x \le b$$

is called a solution for the given problem.
If $y \mapsto \phi(x,y)$ satisfies Lipschitz condition for any $a \le x \le b$,
then the given problem has at most one solution.

Existence.

Thm.

Let f be a real-valued continuous bounded function on $[0,1]\times\mathbb{R}$.

Then, the initial-value problem

$$\begin{cases} y' = f(x,y) \\ y(0) = C \in \mathbb{R} \end{cases}$$

has a solution.

Proof